

# AP<sup>®</sup> Physics C: Electricity and Magnetism Practice Exam

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From the 2016 Administration

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# ADVANCED PLACEMENT PHYSICS C TABLE OF INFORMATION

| CONSTANTS AND CONVERSION FACTORS                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proton mass, $m_p = 1.67 \times 10^{-27}$ kg<br>Neutron mass, $m_n = 1.67 \times 10^{-27}$ kg<br>Electron mass, $m_e = 9.11 \times 10^{-31}$ kg<br>Avogadro's number, $N_0 = 6.02 \times 10^{23}$ mol <sup>-1</sup><br>Universal gas constant, $R = 8.31$ J/(mol·K)<br>Boltzmann's constant, $k_B = 1.38 \times 10^{-23}$ J/K | Electron charge magnitude, $e = 1.60 \times 10^{-19}$ C<br>1 electron volt, $1 \text{ eV} = 1.60 \times 10^{-19}$ J<br>Speed of light, $c = 3.00 \times 10^8$ m/s<br>Universal gravitational constant, $G = 6.67 \times 10^{-11}$ (N·m <sup>2</sup> )/kg <sup>2</sup><br>Acceleration due to gravity at Earth's surface, $g = 9.8$ m/s <sup>2</sup>                                             |
| 1 unified atomic mass unit,<br>Planck's constant,<br>Vacuum permittivity,<br>Coulomb's law constant, $k = 1/(4\pi\epsilon_0) = 9.0 \times 10^9$ (N·m <sup>2</sup> )/C <sup>2</sup><br>Vacuum permeability,<br>Magnetic constant, $k' = \mu_0/(4\pi) = 1 \times 10^{-7}$ (T·m)/A<br>1 atmosphere pressure,                     | $1 \text{ u} = 1.66 \times 10^{-27}$ kg = 931 MeV/c <sup>2</sup><br>$h = 6.63 \times 10^{-34}$ J·s = $4.14 \times 10^{-15}$ eV·s<br>$hc = 1.99 \times 10^{-25}$ J·m = $1.24 \times 10^3$ eV·nm<br>$\epsilon_0 = 8.85 \times 10^{-12}$ C <sup>2</sup> /(N·m <sup>2</sup> )<br>$\mu_0 = 4\pi \times 10^{-7}$ (T·m)/A<br>$1 \text{ atm} = 1.0 \times 10^5$ N/m <sup>2</sup> = $1.0 \times 10^5$ Pa |

| UNIT<br>SYMBOLS | meter, m     | mole, mol  | watt, W    | farad, F           |
|-----------------|--------------|------------|------------|--------------------|
|                 | kilogram, kg | hertz, Hz  | coulomb, C | tesla, T           |
|                 | second, s    | newton, N  | volt, V    | degree Celsius, °C |
|                 | ampere, A    | pascal, Pa | ohm, Ω     | electron volt, eV  |
|                 | kelvin, K    | joule, J   | henry, H   |                    |

| PREFIXES          |        |        |
|-------------------|--------|--------|
| Factor            | Prefix | Symbol |
| 10 <sup>9</sup>   | giga   | G      |
| 10 <sup>6</sup>   | mega   | M      |
| 10 <sup>3</sup>   | kilo   | k      |
| 10 <sup>-2</sup>  | centi  | c      |
| 10 <sup>-3</sup>  | milli  | m      |
| 10 <sup>-6</sup>  | micro  | μ      |
| 10 <sup>-9</sup>  | nano   | n      |
| 10 <sup>-12</sup> | pico   | p      |

| VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES |    |              |     |              |     |              |     |
|-----------------------------------------------------|----|--------------|-----|--------------|-----|--------------|-----|
| $\theta$                                            | 0° | 30°          | 37° | 45°          | 53° | 60°          | 90° |
| $\sin \theta$                                       | 0  | 1/2          | 3/5 | $\sqrt{2}/2$ | 4/5 | $\sqrt{3}/2$ | 1   |
| $\cos \theta$                                       | 1  | $\sqrt{3}/2$ | 4/5 | $\sqrt{2}/2$ | 3/5 | 1/2          | 0   |
| $\tan \theta$                                       | 0  | $\sqrt{3}/3$ | 3/4 | 1            | 4/3 | $\sqrt{3}$   | ∞   |

The following assumptions are used in this exam.

- I. The frame of reference of any problem is inertial unless otherwise stated.
- II. The direction of current is the direction in which positive charges would drift.
- III. The electric potential is zero at an infinite distance from an isolated point charge.
- IV. All batteries and meters are ideal unless otherwise stated.
- V. Edge effects for the electric field of a parallel plate capacitor are negligible unless otherwise stated.

# ADVANCED PLACEMENT PHYSICS C EQUATIONS

## MECHANICS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$\vec{J} = \int \vec{F} dt = \Delta \vec{p}$$

$$\vec{p} = m\vec{v}$$

$$|\vec{F}_f| \leq \mu |\vec{F}_N|$$

$$\Delta E = W = \int \vec{F} \cdot d\vec{r}$$

$$K = \frac{1}{2} m v^2$$

$$P = \frac{dE}{dt}$$

$$P = \vec{F} \cdot \vec{v}$$

$$\Delta U_g = mg \Delta h$$

$$a_c = \frac{v^2}{r} = \omega^2 r$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\alpha} = \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_{net}}{I}$$

$$I = \int r^2 dm = \sum m r^2$$

$$x_{cm} = \frac{\sum m_i x_i}{\sum m_i}$$

$$v = r \omega$$

$$\vec{L} = \vec{r} \times \vec{p} = I \vec{\omega}$$

$$K = \frac{1}{2} I \omega^2$$

$$\omega = \omega_0 + \alpha t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$a = \text{acceleration}$$

$$E = \text{energy}$$

$$F = \text{force}$$

$$f = \text{frequency}$$

$$h = \text{height}$$

$$I = \text{rotational inertia}$$

$$J = \text{impulse}$$

$$K = \text{kinetic energy}$$

$$k = \text{spring constant}$$

$$\ell = \text{length}$$

$$L = \text{angular momentum}$$

$$m = \text{mass}$$

$$P = \text{power}$$

$$p = \text{momentum}$$

$$r = \text{radius or distance}$$

$$T = \text{period}$$

$$t = \text{time}$$

$$U = \text{potential energy}$$

$$v = \text{velocity or speed}$$

$$W = \text{work done on a system}$$

$$x = \text{position}$$

$$\mu = \text{coefficient of friction}$$

$$\theta = \text{angle}$$

$$\tau = \text{torque}$$

$$\omega = \text{angular speed}$$

$$\alpha = \text{angular acceleration}$$

$$\phi = \text{phase angle}$$

$$\vec{F}_s = -k \Delta \vec{x}$$

$$U_s = \frac{1}{2} k (\Delta x)^2$$

$$x = x_{\max} \cos(\omega t + \phi)$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$T_s = 2\pi \sqrt{\frac{m}{k}}$$

$$T_p = 2\pi \sqrt{\frac{\ell}{g}}$$

$$|\vec{F}_G| = \frac{G m_1 m_2}{r^2}$$

$$U_G = -\frac{G m_1 m_2}{r}$$

## ELECTRICITY AND MAGNETISM

$$|\vec{F}_E| = \frac{1}{4\pi\epsilon_0} \left| \frac{q_1 q_2}{r^2} \right|$$

$$\vec{E} = \frac{\vec{F}_E}{q}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$E_x = -\frac{dV}{dx}$$

$$\Delta V = -\int \vec{E} \cdot d\vec{r}$$

$$V = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}$$

$$U_E = qV = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$\Delta V = \frac{Q}{C}$$

$$C = \frac{\kappa \epsilon_0 A}{d}$$

$$C_p = \sum_i C_i$$

$$\frac{1}{C_s} = \sum_i \frac{1}{C_i}$$

$$I = \frac{dQ}{dt}$$

$$U_C = \frac{1}{2} Q \Delta V = \frac{1}{2} C (\Delta V)^2$$

$$R = \frac{\rho \ell}{A}$$

$$\vec{E} = \rho \vec{J}$$

$$I = Ne v_d A$$

$$I = \frac{\Delta V}{R}$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

$$P = I \Delta V$$

$$A = \text{area}$$

$$B = \text{magnetic field}$$

$$C = \text{capacitance}$$

$$d = \text{distance}$$

$$E = \text{electric field}$$

$$\mathcal{E} = \text{emf}$$

$$F = \text{force}$$

$$I = \text{current}$$

$$J = \text{current density}$$

$$L = \text{inductance}$$

$$\ell = \text{length}$$

$$n = \text{number of loops of wire per unit length}$$

$$N = \text{number of charge carriers per unit volume}$$

$$P = \text{power}$$

$$Q = \text{charge}$$

$$q = \text{point charge}$$

$$R = \text{resistance}$$

$$r = \text{radius or distance}$$

$$t = \text{time}$$

$$U = \text{potential or stored energy}$$

$$V = \text{electric potential}$$

$$v = \text{velocity or speed}$$

$$\rho = \text{resistivity}$$

$$\Phi = \text{flux}$$

$$\kappa = \text{dielectric constant}$$

$$\vec{F}_M = q\vec{v} \times \vec{B}$$

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{\ell} \times \hat{r}}{r^2}$$

$$\vec{F} = \int I d\vec{\ell} \times \vec{B}$$

$$B_s = \mu_0 n I$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$$

$$\mathcal{E} = -L \frac{dI}{dt}$$

$$U_L = \frac{1}{2} L I^2$$

# ADVANCED PLACEMENT PHYSICS C EQUATIONS

## GEOMETRY AND TRIGONOMETRY

Rectangle

$$A = bh$$

Triangle

$$A = \frac{1}{2}bh$$

Circle

$$A = \pi r^2$$

$$C = 2\pi r$$

$$s = r\theta$$

Rectangular Solid

$$V = \ell wh$$

Cylinder

$$V = \pi r^2 \ell$$

$$S = 2\pi r \ell + 2\pi r^2$$

Sphere

$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2$$

Right Triangle

$$a^2 + b^2 = c^2$$

$$\sin \theta = \frac{a}{c}$$

$$\cos \theta = \frac{b}{c}$$

$$\tan \theta = \frac{a}{b}$$

$A$  = area

$C$  = circumference

$V$  = volume

$S$  = surface area

$b$  = base

$h$  = height

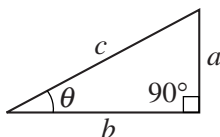
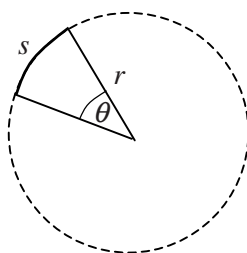
$\ell$  = length

$w$  = width

$r$  = radius

$s$  = arc length

$\theta$  = angle



## CALCULUS

$$\frac{df}{dx} = \frac{df}{du} \frac{du}{dx}$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(\ln ax) = \frac{1}{x}$$

$$\frac{d}{dx}[\sin(ax)] = a \cos(ax)$$

$$\frac{d}{dx}[\cos(ax)] = -a \sin(ax)$$

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}$$

$$\int \frac{dx}{x+a} = \ln|x+a|$$

$$\int \cos(ax) dx = \frac{1}{a} \sin(ax)$$

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax)$$

## VECTOR PRODUCTS

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$|\vec{A} \times \vec{B}| = AB \sin \theta$$

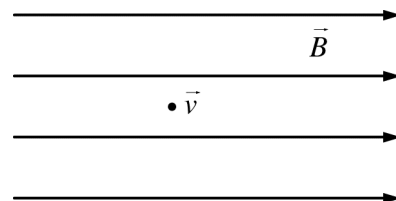
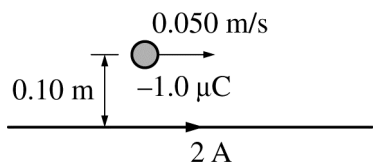
# PHYSICS C: ELECTRICITY AND MAGNETISM

## SECTION I

Time—45 minutes

35 Questions

**Directions:** Each of the questions or incomplete statements below is followed by five suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.



1. A small object of charge  $-1.0 \mu\text{C}$  that is  $0.10 \text{ m}$  above a long, straight wire moves at a speed of  $0.050 \text{ m/s}$  parallel to the wire, as shown in the figure above. The current in the wire is  $2 \text{ A}$ . What is the magnitude and direction of the magnetic force on the object?
  - (A)  $2 \times 10^{-13} \text{ N}$ , toward the wire
  - (B)  $2 \times 10^{-13} \text{ N}$ , away from the wire
  - (C)  $4\pi \times 10^{-13} \text{ N}$ , toward the wire
  - (D)  $4\pi \times 10^{-13} \text{ N}$ , away from the wire
  - (E)  $0$ , direction is undefined
2. An electron travels through a uniform magnetic field  $\vec{B}$  directed to the right. The electron's velocity  $\vec{v}$  is directed out of the page, as shown in the figure above. The direction of the force on the electron due to the magnetic field at the instant shown is
  - (A) toward the top of the page
  - (B) toward the bottom of the page
  - (C) to the left
  - (D) to the right
  - (E) into the page
3. Two long parallel wires have currents of  $I_1$  and  $I_2$  and are a distance  $d$  apart. The force per unit length exerted by one wire on the other is  $F/L$ . The current in each wire is doubled, and the distance between the wires is reduced to  $d/2$ . The new force per unit length is
  - (A)  $\frac{F}{2L}$
  - (B)  $\frac{F}{L}$
  - (C)  $\frac{2F}{L}$
  - (D)  $\frac{4F}{L}$
  - (E)  $\frac{8F}{L}$

### Questions 4-5

A capacitor is constructed of two large, identical, parallel metal plates separated by a small distance  $d$ . A battery fully charges the capacitor and is then disconnected.

4. The plate separation is now increased to a distance  $2d$ . Which of the following correctly describes the change, if any, of the voltage across the capacitor, the electric field between the plates, and the energy stored in the capacitor?

| <u>Voltage</u>      | <u>Electric Field</u> | <u>Energy</u>   |
|---------------------|-----------------------|-----------------|
| (A) Does not change | Does not change       | Does not change |
| (B) Does not change | Doubles               | Does not change |
| (C) Doubles         | Doubles               | Does not change |
| (D) Doubles         | Does not change       | Doubles         |
| (E) Doubles         | Doubles               | Doubles         |

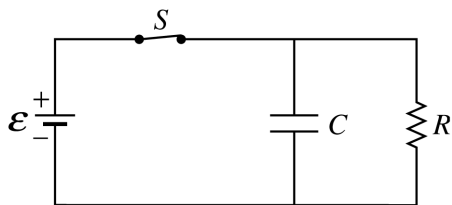
5. If, instead of separating the plates, the empty space between the plates is filled with a slab of insulating material that has a dielectric constant  $K = 2$ , which of the following correctly describes the change, if any, of the voltage across the capacitor, the electric field between the plates, and the energy stored in the capacitor?

| <u>Voltage</u>      | <u>Electric Field</u> | <u>Energy</u>   |
|---------------------|-----------------------|-----------------|
| (A) Halves          | Halves                | Halves          |
| (B) Halves          | Halves                | Does not change |
| (C) Does not change | Does not change       | Halves          |
| (D) Does not change | Halves                | Halves          |
| (E) Does not change | Does not change       | Does not change |

- 
6. A capacitor stores energy  $U_1$  when it holds charge  $Q$ . The same capacitor stores energy  $U_2$  when it holds charge  $16Q$ . What is the ratio  $U_2/U_1$ ?

- (A) 8  
(B) 16  
(C) 64  
(D) 128  
(E) 256

### Questions 7-8



Switch  $S$  in the circuit shown above has been closed for a long time.

7. Which of the following is the correct expression for the current in the resistor?

- (A)  $\frac{\mathcal{E}}{R}$   
 (B)  $\frac{C\mathcal{E}}{R}$   
 (C)  $C\mathcal{E}$   
 (D)  $\frac{\mathcal{E}}{RC}$   
 (E) Zero

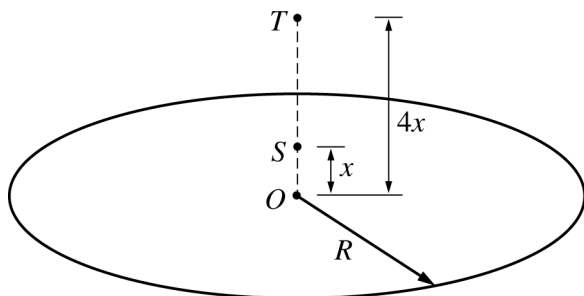
8. Suppose that the switch is opened at time  $t = 0$ . Which of the following combinations of a differential equation and an initial condition can be used to solve for the charge  $Q(t)$  on the upper plate of the capacitor as a function of time  $t$ ?

- | <u>Differential Equation</u>                     | <u>Initial Condition</u> |
|--------------------------------------------------|--------------------------|
| (A) $\frac{Q}{C} - R\frac{dQ}{dt} = 0$           | $Q(0) = C\mathcal{E}$    |
| (B) $\frac{Q}{C} - R\frac{dQ}{dt} = \mathcal{E}$ | $Q(0) = 0$               |
| (C) $\frac{Q}{C} + R\frac{dQ}{dt} = \mathcal{E}$ | $Q(0) = C\mathcal{E}$    |
| (D) $\frac{Q}{C} + R\frac{dQ}{dt} = 0$           | $Q(0) = 0$               |
| (E) $\frac{Q}{C} + R\frac{dQ}{dt} = 0$           | $Q(0) = C\mathcal{E}$    |

9. A nonconducting sphere of radius  $R$  has a uniform volume charge density  $\rho$ . At the surface of the sphere, the electric field strength is  $E$ . If a second sphere of radius  $2R$  was created of the same material and with the same charge density, what would be the strength of the electric field at the surface of the second sphere?

- (A)  $E/4$   
 (B)  $E/2$   
 (C)  $E$   
 (D)  $2E$   
 (E)  $4E$

### Questions 10-11



Note: Figure not drawn to scale.

The figure above shows a thin, circular nonconducting sheet of positive charge uniformly distributed over its area. The radius of the sheet is  $R$ . Point  $O$  is at the center of the sheet. Point  $S$  is a distance  $x$  from the center of the sheet, and point  $T$  is a distance  $4x$  from the center of the sheet. Assume  $R \gg x$ .

10. If the magnitude of the electric field at point  $T$  is  $E_T$ , which of the following best represents the magnitude of the electric field at point  $S$ ?

(A)  $E_T/16$   
 (B)  $E_T/4$   
 (C)  $E_T$   
 (D)  $4E_T$   
 (E)  $16E_T$

11. A small sphere of mass  $m$  and charge  $-q$  is released from rest at point  $T$ . If the electric potentials at points  $S$  and  $T$  are  $V_S$  and  $V_T$ , respectively, what is the speed of the sphere when it reaches point  $S$ ? Ignore the effects of gravity.

(A)  $\frac{2q}{m}(V_S + V_T)$   
 (B)  $\frac{4q}{m}(V_S + V_T)$   
 (C)  $\sqrt{\frac{q}{2m}}(V_S - V_T)$   
 (D)  $\sqrt{\frac{q}{2m}}(V_S + V_T)$   
 (E)  $\sqrt{\frac{2q}{m}}(V_S - V_T)$



# Questions 12-13

| Capacitor        | $K$ | $L$  | $M$  | $N$  |
|------------------|-----|------|------|------|
| Plate area       | $A$ | $A$  | $2A$ | $2A$ |
| Plate separation | $d$ | $2d$ | $d$  | $2d$ |

The chart above shows the plate area and separation for four different air-filled parallel-plate capacitors:  $K$ ,  $L$ ,  $M$ , and  $N$ .

12. Each capacitor is charged such that the same electric field is created between the plates. Which of the following correctly ranks the charge stored on each capacitor?

- (A)  $(Q_M = Q_N) > (Q_K = Q_L)$
- (B)  $Q_M > Q_K > Q_N > Q_L$
- (C)  $(Q_K = Q_M) > (Q_L = Q_N)$
- (D)  $Q_K = Q_L = Q_M = Q_N$
- (E)  $Q_M > (Q_K = Q_N) > Q_L$

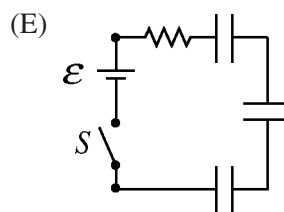
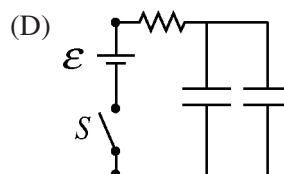
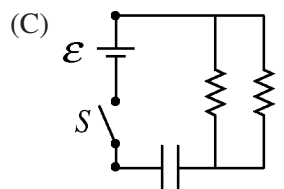
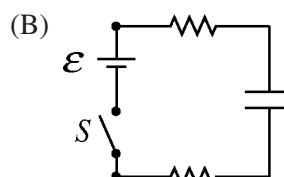
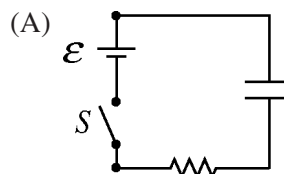
13. Each capacitor is charged such that the same potential difference exists across the plates. Which of the following correctly ranks the magnitude of electric field between the plates of each capacitor?

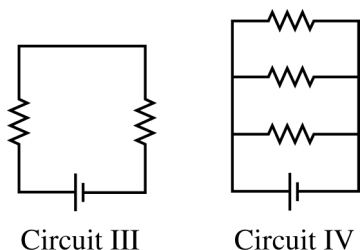
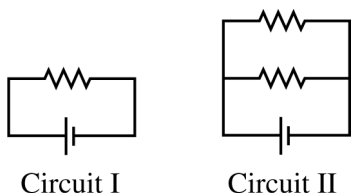
- (A)  $(E_M = E_N) > (E_K = E_L)$
- (B)  $E_M > E_K > E_N > E_L$
- (C)  $(E_K = E_M) > (E_L = E_N)$
- (D)  $E_K = E_L = E_M = E_N$
- (E)  $E_L > (E_K = E_N) > E_M$

14. A parallel-plate capacitor stores 240 nC when fully charged by the application of a 12 V potential difference across its plates. What is its capacitance?

- (A)  $5.0 \times 10^{-11}$  F
- (B)  $1.7 \times 10^{-9}$  F
- (C)  $2.0 \times 10^{-8}$  F
- (D)  $2.9 \times 10^{-6}$  F
- (E)  $4.8 \times 10^{-6}$  F

15. Identical resistors and identical uncharged capacitors are connected to identical ideal batteries of emf  $\mathcal{E}$  in the circuits shown below. Each circuit has a switch  $S$  in the open position. In which circuit will the capacitors reach half their maximum charge in the least amount of time when switch  $S$  is closed?



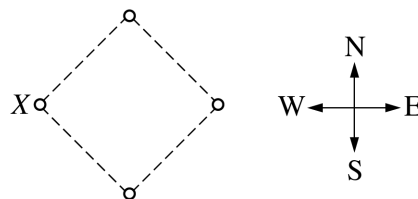


16. All four circuits shown above contain identical ideal batteries and identical resistors. Which circuits draw the most and least power from the battery?

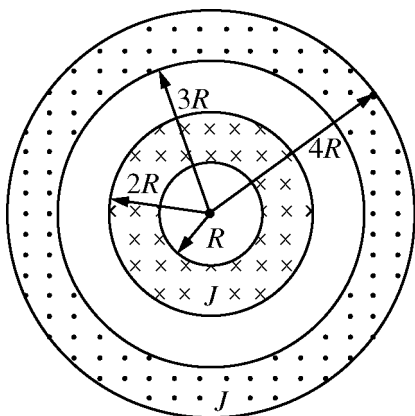
|     | <u>Most Power</u> | <u>Least Power</u> |
|-----|-------------------|--------------------|
| (A) | I                 | II                 |
| (B) | I                 | V                  |
| (C) | II                | III                |
| (D) | IV                | I                  |
| (E) | IV                | III                |

17. The current in a  $10\ \Omega$  resistor is given as a function of time  $t$  by the equation  $I = I_0 e^{-\alpha t}$ , where  $I_0 = 4\ \text{A}$  and  $\alpha = 2\ \text{s}^{-1}$ . What is the total energy dissipated in the resistor from time  $t = 0$  until the current becomes zero?

- (A) 20 J  
(B) 40 J  
(C) 80 J  
(D) 160 J  
(E) 320 J

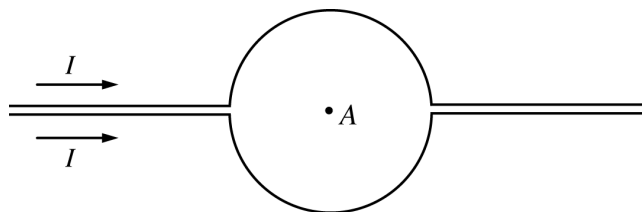


18. Four conducting wires perpendicular to the plane of the page are at the corners of a square, as shown in cross section in the figure above. The wires carry equal currents in the same direction. What is the direction of the total magnetic force on wire  $X$  caused by the other three wires?
- (A) To the north  
(B) To the south  
(C) To the east  
(D) To the west  
(E) In none of the directions above, since the magnitude of the force is zero



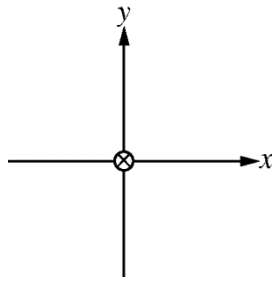
19. Two long, hollow, concentric conducting cylinders carry currents in opposite directions into and out of the plane of the page, as shown in the cross section above. The currents are unequal, but the current density  $J$  is the same for both cylinders. In which of the following regions can the net magnetic field be zero at some nonzero finite distance  $r$  from the central axis?

(A)  $r < R$  only  
 (B) Both  $r < R$  and  $R < r < 2R$   
 (C) Both  $r < R$  and  $2R < r < 3R$   
 (D) Both  $r < R$  and  $3R < r < 4R$   
 (E)  $r > 4R$  only

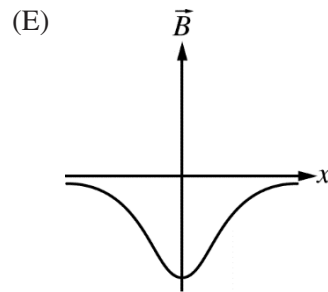
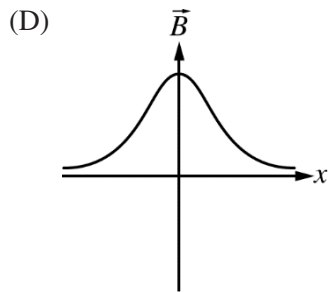
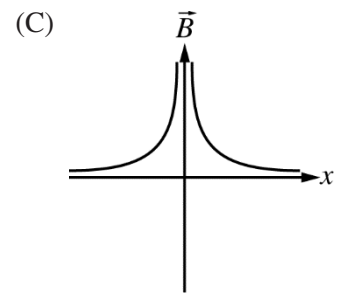
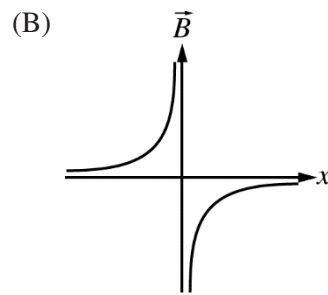
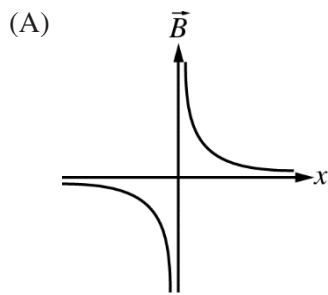


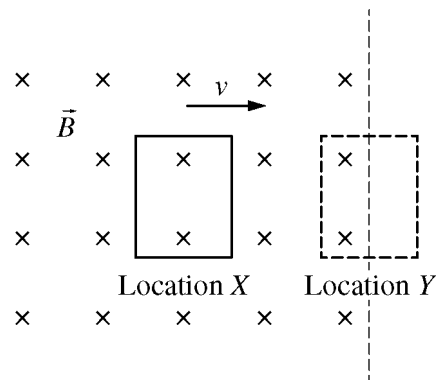
20. The middle sections of two long, straight wires are each bent into semicircular shapes. The two wires are placed close to each other in a plane as shown above. Point A is the center of the circle formed by the two wires. There is a current  $I$  in each wire. Which of the following best represents the direction of the magnetic field, if any, at point A due to the currents?

(A)   
 (B)   
 (C)  (Out of the page)  
 (D)  (Into the page)  
 (E) There is no direction, because the magnetic field is zero.



21. A long straight wire is located at the origin of an  $xy$ -coordinate system and is perpendicular to the  $xy$ -plane. The wire has a current that is directed into the page, as shown in the figure above. Which of the following graphs best represents the magnetic field  $\vec{B}$  due to the current as a function of the position  $x$  along the  $x$ -axis? Assume a positive magnetic field to be directed in the  $+y$  direction.





22. A loop of wire at location  $X$  is moving toward the right with constant speed  $v$  in a region of uniform magnetic field  $\vec{B}$ , which is perpendicular to the plane of the loop. The magnetic field region ends at the dashed line, as shown in the figure above. Later, the loop is at location  $Y$  and exiting the magnetic field with the same constant speed  $v$ . The process is then repeated with the loop moving at a speed of  $2v$ . Which of the following best describes the emf in the loop at the two positions shown when the process is repeated at a speed of  $2v$ ?

Emf at Location  $X$

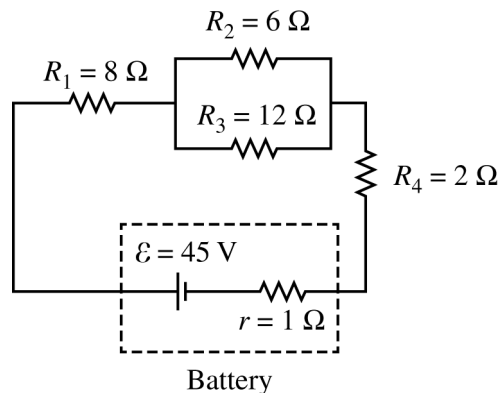
- (A) Nonzero and halved
- (B) Nonzero and doubled
- (C) Nonzero and doubled
- (D) Zero at both speeds
- (E) Zero at both speeds

Emf at Location  $Y$

- Nonzero and halved
- Nonzero and doubled
- Zero at both speeds
- Zero at both speeds
- Nonzero and doubled

23. A long solenoid has 300 loops. The region inside the solenoid has a uniform magnetic field that is directed parallel to the axis of the solenoid. Each loop has an area  $2 \times 10^{-4} \text{ m}^2$ . The magnetic field decreases at a constant rate from 0.5 T to zero. What is the time duration for the change in magnetic field if a potential difference of 10 V is induced between the ends of the solenoid while the field is decreasing?

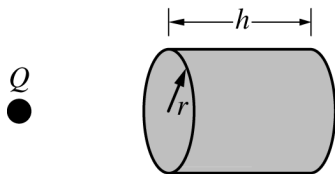
- (A)  $1 \times 10^{-5} \text{ s}$   
 (B)  $3 \times 10^{-5} \text{ s}$   
 (C)  $3 \times 10^{-3} \text{ s}$   
 (D)  $1 \times 10^{-2} \text{ s}$   
 (E)  $3 \times 10^{-2} \text{ s}$



#### Questions 24-25

The circuit represented in the figure above contains four resistors and a battery. The 45 V battery has a  $1 \Omega$  internal resistance.

24. Which of the following ranks the absolute values of the potential differences  $\Delta V$  across the resistors from highest to lowest?
- (A)  $\Delta V_3 > \Delta V_1 > \Delta V_2 > \Delta V_4$   
 (B)  $\Delta V_4 > \Delta V_2 > \Delta V_1 > \Delta V_3$   
 (C)  $\Delta V_1 > \Delta V_4 > \Delta V_2 > \Delta V_3$   
 (D)  $\Delta V_1 > (\Delta V_2 = \Delta V_3) > \Delta V_4$   
 (E)  $\Delta V_4 > (\Delta V_2 = \Delta V_3) > \Delta V_1$
25. How much energy is dissipated by the battery's internal resistance in 60 s?
- (A) 9 J  
 (B) 180 J  
 (C) 540 J  
 (D) 900 J  
 (E) 8100 J



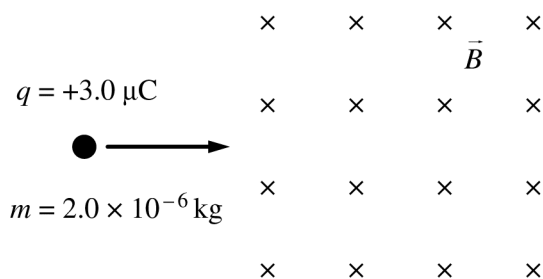
26. A closed cylindrical shell of volume  $\pi r^2 h$  is placed close to an object with a charge of  $Q$ , as shown in the figure above. There are no other charged objects nearby. The electric flux through the closed cylindrical shell is

(A) 0  
 (B)  $Q/\epsilon_0$   
 (C)  $\pi r^2 Q$   
 (D)  $2\pi r h Q$   
 (E)  $2\pi r^2 Q + 2\pi r h Q$

27. The electric field is zero everywhere within a certain region of space. The electric potential everywhere within the region
- (A) must be zero  
 (B) must be uniform  
 (C) must be positive  
 (D) must be negative  
 (E) can be both positive and negative in different parts of the region

28. Two particles with positive charges  $q_1$  and  $q_2$  are both at rest and are very far apart. How much work would be done by external forces to bring the particles to rest at a distance  $d$  apart?

(A) Zero  
 (B)  $\frac{kq_1q_2}{d}$   
 (C)  $-\frac{kq_1q_2}{d}$   
 (D)  $\frac{2kq_1q_2}{d}$   
 (E)  $-\frac{2kq_1q_2}{d}$

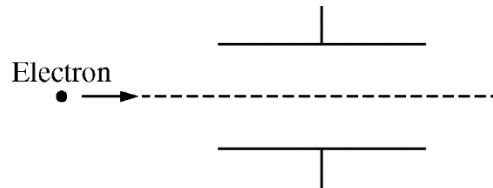


29. A small object with a charge of  $q = +3.0 \mu\text{C}$  and a mass  $m = 2.0 \times 10^{-6} \text{ kg}$  enters a magnetic field of magnitude  $B = 0.20 \text{ T}$  directed into the page, as shown in the figure above. If the speed of the object is  $1000 \text{ m/s}$ , the object's acceleration at the moment it enters the field is most nearly

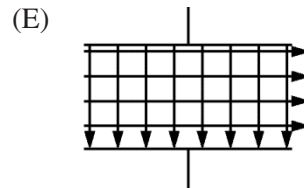
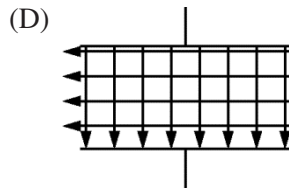
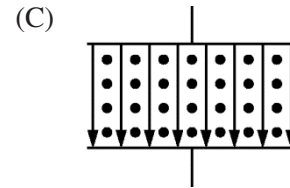
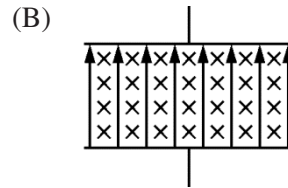
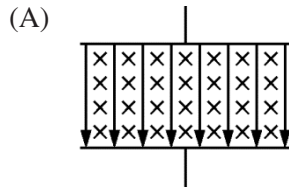
(A) zero because the velocity is perpendicular to the magnetic field  
 (B)  $300 \text{ m/s}^2$  toward the bottom of the page  
 (C)  $300 \text{ m/s}^2$  toward the top of the page  
 (D)  $600 \text{ m/s}^2$  toward the bottom of the page  
 (E)  $600 \text{ m/s}^2$  toward the top of the page

30. A  $0.20 \text{ m}$  long wire carries a current of  $15 \text{ A}$  and lies perpendicular to a magnetic field. The magnetic force on the wire is measured to be  $0.060 \text{ N}$ . What is the magnitude of the magnetic field?

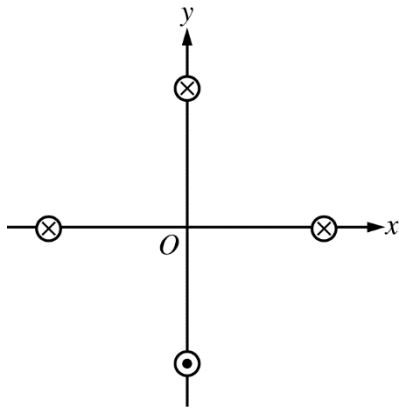
(A)  $0.010 \text{ T}$   
 (B)  $0.020 \text{ T}$   
 (C)  $0.040 \text{ T}$   
 (D)  $0.060 \text{ T}$   
 (E)  $0.080 \text{ T}$



31. Uniform magnetic and electric fields exist between the two oppositely charged parallel plates shown in the figure above. An electron travels horizontally between the plates. Assuming gravitational effects to be negligible, which of the following diagrams shows a combination of electric and magnetic field directions that will allow the electron to travel undeflected?







32. Four long, straight wires are perpendicular to the  $xy$ -plane. Each wire is the same distance from the origin  $O$ , as shown in the figure above. The wires have equal currents that are in the directions shown. Which of the following best represents the direction of the net magnetic field, if any, at the origin due to the four currents?

- (A) 
- (B) 
- (C) 
- (D) 

(E) Undefined because the net field is zero.

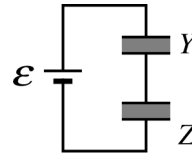


Figure 1

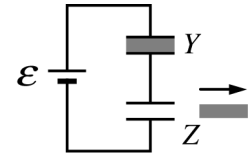
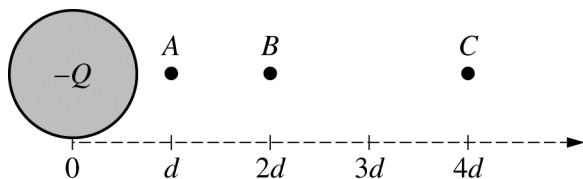


Figure 2

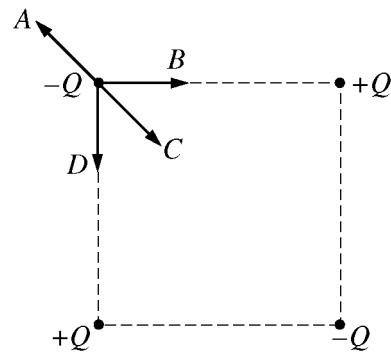
33. Two identical capacitors, Y and Z, are connected in series with an ideal battery, as shown in figure 1 above, and fully charged. Each capacitor has a dielectric slab of dielectric constant  $\kappa > 1$  between its plates. If the dielectric slab is removed from capacitor Z, which of the following describes what happens to the voltage across each capacitor?

| Voltage across<br>Capacitor Y | Voltage across<br>Capacitor Z |
|-------------------------------|-------------------------------|
| (A) Increases                 | Decreases                     |
| (B) Increases                 | Increases                     |
| (C) Remains the same          | Remains the same              |
| (D) Decreases                 | Decreases                     |
| (E) Decreases                 | Increases                     |



34. A particle of charge  $+q$  is released from rest at position  $B$ , which is a distance  $2d$  from the center of a fixed nonconducting sphere that has a charge of  $-Q$  distributed uniformly throughout its volume, as shown in the figure above. When the particle reaches position  $A$ , which is a distance  $d$  from the center of the charged sphere, its kinetic energy is  $K_0$ . The same particle is now released from rest at position  $C$ , which is a distance  $4d$  from the center of the nonconducting sphere. The kinetic energy of the particle when it again reaches position  $A$  is

- (A)  $K_0$
- (B)  $3K_0/2$
- (C)  $2K_0$
- (D)  $5K_0/2$
- (E)  $3K_0$



35. Four point charges of equal magnitude but different signs are arranged on the corners of a square as shown above. Which of the vectors shown represents the direction of the net force acting on the charge at the upper left-hand corner of the square due to the other charges?
- (A)  $A$
  - (B)  $B$
  - (C)  $C$
  - (D)  $D$
  - (E) It has no direction because the force is zero.

## S T O P

### END OF ELECTRICITY AND MAGNETISM SECTION I

IF YOU FINISH BEFORE TIME IS CALLED,  
YOU MAY CHECK YOUR WORK ON ELECTRICITY AND MAGNETISM SECTION I ONLY.

DO NOT TURN TO ANY OTHER TEST MATERIALS.

**MAKE SURE YOU HAVE DONE THE FOLLOWING.**

- **PLACED YOUR AP NUMBER LABEL ON YOUR ANSWER SHEET**
- **WRITTEN AND GRIDDED YOUR AP NUMBER CORRECTLY ON YOUR ANSWER SHEET**
- **TAKEN THE AP EXAM LABEL FROM THE FRONT OF THIS BOOKLET AND PLACED IT ON YOUR ANSWER SHEET**

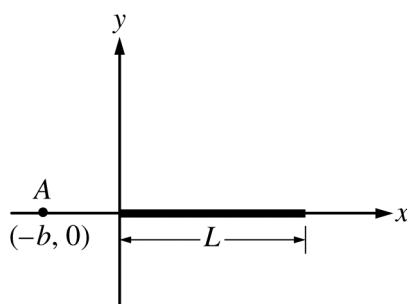
**PHYSICS C: ELECTRICITY AND MAGNETISM**

**SECTION II**

**Time—45 minutes**

**3 Questions**

**Directions:** Answer all three questions. The suggested time is about 15 minutes for answering each of the questions, which are worth 15 points each. The parts within a question may not have equal weight. Show all your work in this booklet in the spaces provided after each part.



E&M.1.

A rod of length  $L$  lies along the  $x$ -axis with its left end at the origin, as shown in the figure above. The rod has a nonuniform linear charge density  $\lambda = \alpha x$ , where  $\alpha$  is a positive constant.

- (a) Determine the units of the constant  $\alpha$ .
- (b) Derive an expression for the total charge on the rod. Express your answer in terms of  $\alpha$ ,  $L$ , and physical constants, as appropriate.

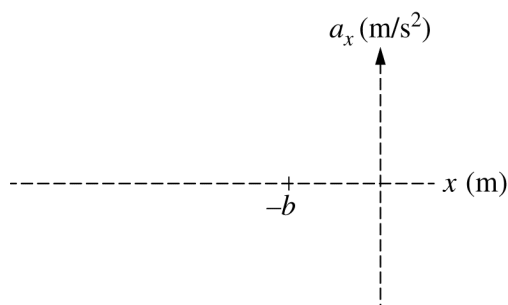
The rod is replaced with one of the same length but with a uniform positive linear charge density  $\lambda_0$ .

(c) Show that the electric field at point A, which is a distance  $b$  to the left of the rod, has a magnitude of

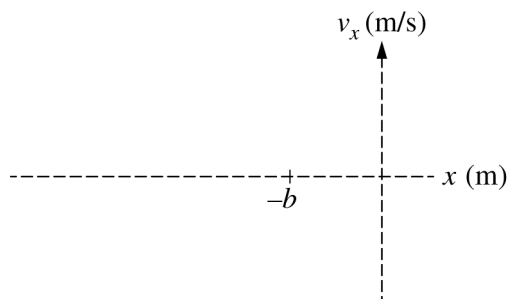
$$E = \frac{\lambda_0 L}{4\pi\epsilon_0 b(b + L)}.$$

(d) A proton is placed at point A, at position  $(-b, 0)$ , near the positively charged rod and released from rest.

- i. Sketch a graph of the horizontal component of the acceleration of the proton  $a_x$  as a function of its position  $x$  for the region in which the proton travels after it is released. Sketch any asymptote, but do not indicate its value.



- ii. Sketch a graph of the horizontal component of the velocity of the proton  $v_x$  as a function of its position  $x$  for the region in which the proton travels after it is released. Sketch any asymptote but do not indicate its value.

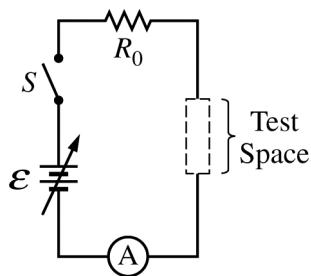


Question 1 continues on next page.

- iii. Instead of a proton, an alpha particle (a particle with twice the charge and four times the mass of a proton) is now placed at point  $A$ . Is the magnitude of the initial acceleration of the alpha particle greater than, less than, or equal to the magnitude of the initial acceleration of the proton from part (d)i?

\_\_\_\_\_ Greater than      \_\_\_\_\_ Less than      \_\_\_\_\_ Equal to

- (e) Using the equation from part (c), show that the magnitude of the electric field at a point on the  $x$ -axis very far away from the rod is approximately equal to the electric field due to a point charge of magnitude  $Q = \lambda_0 L$  located at the origin.



E&M.2.

The circuit above consists of a variable DC power supply, an ammeter  $A$ , a switch  $S$ , and a resistor of unknown resistance  $R_0$  connected in series with a test space into which different circuit elements can be placed. In order to determine  $R_0$ , a wire of negligible resistance is placed in the test space. The power supply is set at various potential differences, and the current in the circuit is measured. The data are shown below.

|                   |       |       |       |       |
|-------------------|-------|-------|-------|-------|
| $\mathcal{E}$ (V) | 2.00  | 4.00  | 6.00  | 8.00  |
| Current (mA)      | 0.146 | 0.310 | 0.433 | 0.577 |
|                   |       |       |       |       |
|                   |       |       |       |       |

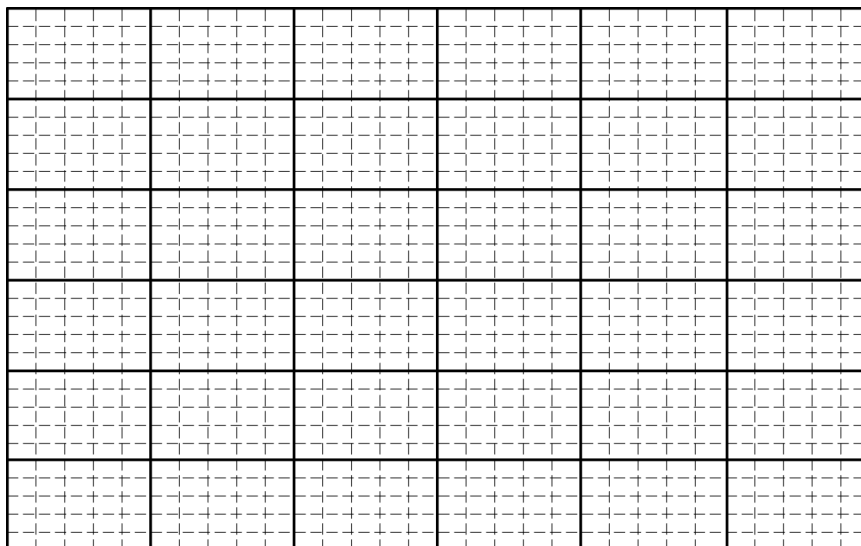
- (a) Indicate below which quantities should be graphed to yield a straight line that has a slope that could be used to calculate a numerical value for the resistance  $R_0$ .

Horizontal axis: \_\_\_\_\_

Vertical axis: \_\_\_\_\_

Use the remaining rows in the table above, as needed, to record any quantities that you indicated that are not given.

- (b) Plot the straight line data points on the graph below. Clearly scale and label all axes, including units, if appropriate. Draw a straight line that best represents the data.

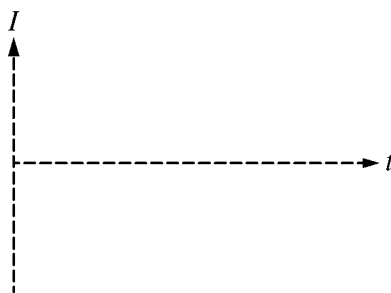


- (c) Using your straight line, calculate an experimental value for  $R_0$ .

Question 2 continues on next page.

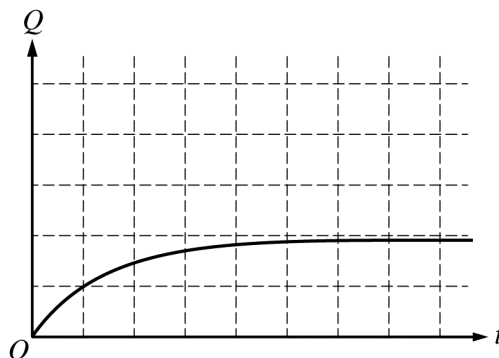
(d) The power supply is set to 4.00 V and an uncharged, air-filled capacitor is placed in the test space. The switch is closed at time  $t = 0$ , and the capacitor is allowed to fully charge.

- i. On the axes shown, sketch the current  $I$  through the ammeter as a function of time  $t$ . Explicitly label any intercepts, asymptotes, maxima, or minima with numerical values or algebraic expressions, as appropriate.

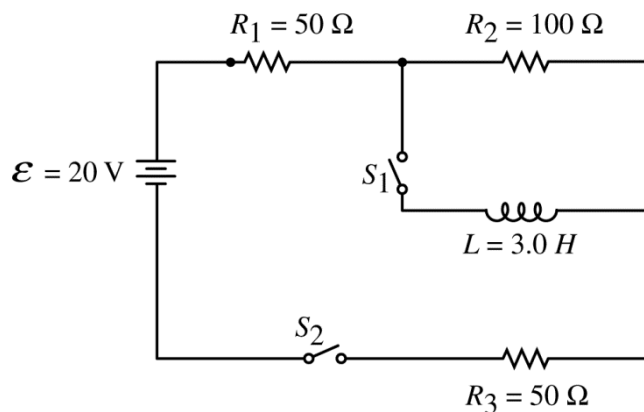


- ii. It is determined that the time constant is 0.045 s. The capacitor is charged to 4 V. Calculate the energy stored in the capacitor.

- iii. The graph below shows the charge  $Q$  on the capacitor as a function of time  $t$ . The capacitor is discharged, and a dielectric with dielectric constant  $\kappa = 2$  is inserted between the plates. The same charging procedure is repeated. On the graph below, sketch the curve showing the charge  $Q$  on the dielectric-filled capacitor as a function of time  $t$ .







E&M.3.

An ideal battery is connected to three resistors, an ideal inductor, and two switches, as shown in the circuit above. Both switches are initially open. Switch  $S_2$  is then closed.

- (a) Calculate the voltage across resistor  $R_2$ .

Switch  $S_1$  is then closed.

- (b) Calculate the magnitude of the rate of change of the current through the inductor immediately after switch  $S_1$  is closed.

- (c) At time  $t$  after switch  $S_1$  is closed, the instantaneous current through resistor  $R_2$  is measured to be 0.020 A.

- i. Calculate the current through resistor  $R_3$  at time  $t$ .

- ii. Calculate the energy stored in the inductor at time  $t$ .

After a long time, the currents reach a steady value. Switch  $S_2$  is then opened.

(d) Calculate the current through the inductor immediately after switch  $S_2$  is opened.

(e) Write, but do NOT solve, a differential equation that can be used to determine the current through resistor  $R_2$  at time  $t$  after switch  $S_2$  is opened.

(f) Using the differential equation from part (e), derive an expression for the current through resistor  $R_2$  as a function of time  $t$  after switch  $S_2$  is opened.

**Answer Key for AP Physics C: Electricity and Magnetism  
Practice Exam, Section I**

|                |                |
|----------------|----------------|
| Question 1: B  | Question 19: D |
| Question 2: B  | Question 20: E |
| Question 3: E  | Question 21: B |
| Question 4: D  | Question 22: E |
| Question 5: A  | Question 23: C |
| Question 6: E  | Question 24: D |
| Question 7: A  | Question 25: C |
| Question 8: E  | Question 26: A |
| Question 9: D  | Question 27: B |
| Question 10: C | Question 28: B |
| Question 11: E | Question 29: C |
| Question 12: A | Question 30: B |
| Question 13: C | Question 31: A |
| Question 14: C | Question 32: D |
| Question 15: E | Question 33: E |
| Question 16: E | Question 34: B |
| Question 17: B | Question 35: C |
| Question 18: C |                |

## 2016 AP Physics C: Electricity and Magnetism Question Descriptors and Performance Data

### Multiple-Choice Questions

| Questions | Topic                             | Key | % Correct |
|-----------|-----------------------------------|-----|-----------|
| 1         | Magnetostatics                    | B   | 40        |
| 2         | Magnetostatics                    | B   | 58        |
| 3         | Magnetostatics                    | E   | 41        |
| 4         | Conductors/Capacitors/Dielectrics | D   | 43        |
| 5         | Conductors/Capacitors/Dielectrics | A   | 50        |
| 6         | Conductors/Capacitors/Dielectrics | E   | 34        |
| 7         | Electric Circuits                 | A   | 75        |
| 8         | Electric Circuits                 | E   | 13        |
| 9         | Electrostatics                    | D   | 29        |
| 10        | Electrostatics                    | C   | 19        |
| 11        | Electrostatics                    | E   | 53        |
| 12        | Conductors/Capacitors/Dielectrics | A   | 19        |
| 13        | Conductors/Capacitors/Dielectrics | C   | 45        |
| 14        | Conductors/Capacitors/Dielectrics | C   | 86        |
| 15        | Electric Circuits                 | E   | 29        |
| 16        | Electric Circuits                 | E   | 70        |
| 17        | Electric Circuits                 | B   | 35        |
| 18        | Magnetostatics                    | C   | 39        |
| 19        | Magnetostatics                    | D   | 28        |
| 20        | Magnetostatics                    | E   | 71        |
| 21        | Magnetostatics                    | B   | 42        |
| 22        | Electromagnetism                  | E   | 55        |
| 23        | Electromagnetism                  | C   | 42        |
| 24        | Electric Circuits                 | D   | 67        |
| 25        | Electric Circuits                 | C   | 43        |
| 26        | Electrostatics                    | A   | 47        |
| 27        | Electrostatics                    | B   | 58        |
| 28        | Electrostatics                    | B   | 48        |
| 29        | Magnetostatics                    | C   | 62        |
| 30        | Magnetostatics                    | B   | 70        |
| 31        | Magnetostatics                    | A   | 52        |
| 32        | Magnetostatics                    | D   | 53        |
| 33        | Conductors/Capacitors/Dielectrics | E   | 36        |
| 34        | Electrostatics                    | B   | 32        |
| 35        | Electrostatics                    | C   | 53        |

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**2016 SCORING GUIDELINES**

**Question 1**

**15 points total**

**Distribution  
of points**

(a) 2 points

For correctly solving the given equation for  $\alpha$

1 point

$$\lambda = \alpha x$$

$$\alpha = \frac{\lambda}{x} = \frac{Q/L}{x}$$

Substituting units into the above equation:

$$\alpha = \frac{\text{C/m}}{\text{m}}$$

For a correct answer

1 point

$$\alpha = \text{C/m}^2$$

(b) 2 points

For integrating with the correct limits or constant of integration

1 point

$$Q = \int \lambda dx = \int_{x=0}^{x=L} \alpha x dx = \left[ \frac{1}{2} \alpha x^2 \right]_{x=0}^{x=L} = \frac{1}{2} \alpha (L^2 - 0)$$

For a correct answer

1 point

$$Q = \frac{1}{2} \alpha L^2$$

(c) 3 points

Using a correct integral expression for Coulomb's law

$$E = \int \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} dq$$

For correctly expressing the charge element in terms of the linear charge density

1 point

$$\lambda_0 = Q/L \text{ so } dq = \lambda_0 dr$$

For correctly substituting into the first equation

1 point

$$E = \int \frac{1}{4\pi\epsilon_0} \frac{\lambda_0}{r^2} dr$$

For integrating with the correct limits or constant of integration

1 point

$$E = \frac{\lambda_0}{4\pi\epsilon_0} \int_{r=b}^{r=b+L} \frac{1}{r^2} dr = \frac{\lambda_0}{4\pi\epsilon_0} \left[ -\frac{1}{r} \right]_{r=b}^{r=b+L} = \frac{\lambda_0}{4\pi\epsilon_0} \left( -\frac{1}{b+L} - \left( -\frac{1}{b} \right) \right)$$

$$E = \frac{\lambda_0}{4\pi\epsilon_0} \left( \frac{-b + b + L}{b(b+L)} \right)$$

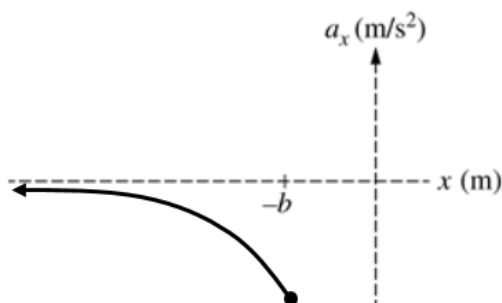
$$E = \frac{1}{4\pi\epsilon_0} \frac{\lambda_0 L}{b(b+L)}$$

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2016 SCORING GUIDELINES**

**Question 1 (continued)**

**Distribution  
of points**

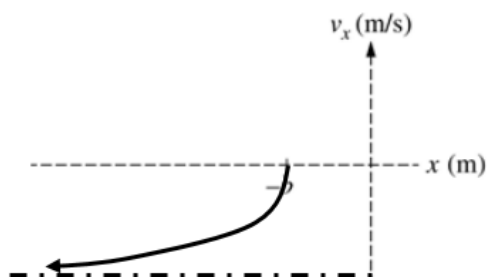
- (d)  
i. 2 points



For a curve starting at negative  $b$ , extending in the negative  $x$  direction, and approaching zero 1 point

For a curve starting at a finite value that is below the horizontal axis and has the horizontal axis as an asymptote 1 point

- ii. 3 points



For a curve starting at negative  $b$  and extending in the negative  $x$  direction 1 point

For a concave up curve that is below the horizontal axis 1 point

For a curve that has a horizontal non-zero asymptote 1 point

- iii. 1 point

For correctly selecting “Less than” 1 point

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**2016 SCORING GUIDELINES**

**Question 1 (continued)**

|                                                                                                 | <b>Distribution<br/>of points</b> |
|-------------------------------------------------------------------------------------------------|-----------------------------------|
| (e) 2 points                                                                                    |                                   |
| For correctly using the assumption that the distance is much greater than the length of the rod | 1 point                           |
| $E = \frac{1}{4\pi\epsilon_0} \frac{\lambda_0 L}{b(b+L)}$                                       |                                   |
| Since $b \gg L$ , $E = \frac{1}{4\pi\epsilon_0} \frac{\lambda_0 L}{b^2}$                        |                                   |
| For correctly substituting $Q$ into the equation above                                          | 1 point                           |
| $Q = \lambda_0 L$                                                                               |                                   |
| $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{b^2}$                                                    |                                   |

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**2016 SCORING GUIDELINES**

**Question 2**

**15 points total**

**Distribution  
of points**

(a) 1 point

For indicating quantities that will yield a straight line that can be used to determine the resistance  $R_0$

1 point

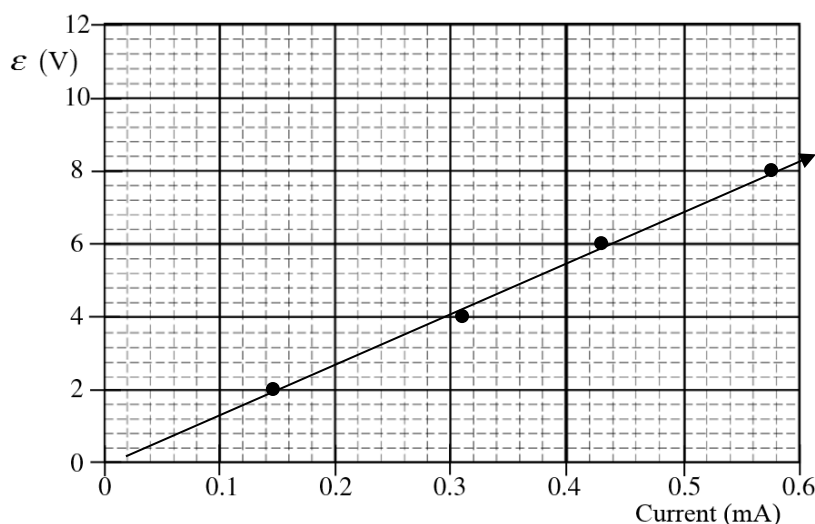
Example:

Horizontal axis: Current

Vertical axis:  $\mathcal{E}$

Note: The axes can be reversed

(b) 4 points



For correctly labeling the axes with appropriate variables and units

1 point

For a correct scale that uses more than half the grid

1 point

For correctly plotting data indicated in part (a)

1 point

For drawing a straight line consistent with the plotted data

1 point

(c) 3 points

For correctly calculating slope using the best-fit straight line and not data points

1 point

$$\text{slope} = \frac{(y_2 - y_1)}{(x_2 - x_1)} = \frac{(8.0 - 2.0) \text{ V}}{(0.58 - 0.15) \text{ mA}} = 13.95 \text{ V/mA}$$

Note: Linear regression gives slope = 14.08 V/mA

To get credit for a linear regression, a response must indicate that the value comes from a linear regression from their calculator and the equation of the line must be shown.

For correctly relating  $R$  to the slope

1 point

$$\mathcal{E} = IR \text{ so } R = \text{slope}$$

For a correct answer

1 point

$$R = 13.95 \text{ k}\Omega$$

Note: Linear regression gives  $R = 14.08 \text{ k}\Omega$

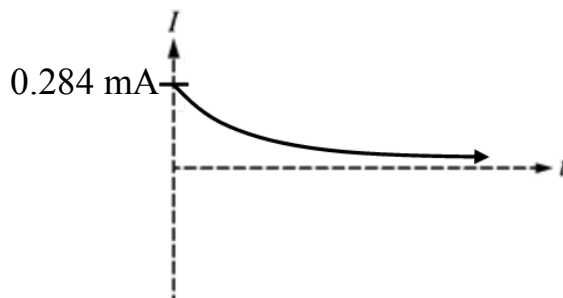


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**2016 SCORING GUIDELINES**

**Question 2 (continued)**

**Distribution  
of points**

- (d)  
i. 3 points



- |                                                                               |         |
|-------------------------------------------------------------------------------|---------|
| For a concave up curve in the first quadrant                                  | 1 point |
| For having the horizontal axis as an asymptote                                | 1 point |
| For correctly labeling the initial current with units or algebraic expression | 1 point |

- ii. 2 points

- |                                                                                                    |         |
|----------------------------------------------------------------------------------------------------|---------|
| For correctly using the time constant to calculate the capacitance of the capacitor<br>$\tau = RC$ | 1 point |
|----------------------------------------------------------------------------------------------------|---------|

$$C = \frac{\tau}{R} = \frac{(0.045 \text{ s})}{(13.95 \times 10^3 \Omega)} = 3.23 \times 10^{-6} \text{ F}$$

Note: Linear regression gives  $C = 3.20 \times 10^{-6} \text{ F}$

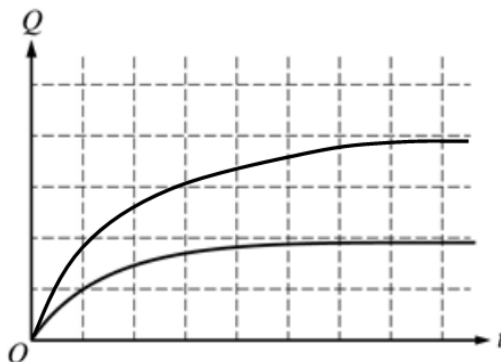
- |                                                                                   |         |
|-----------------------------------------------------------------------------------|---------|
| For a correct application of an expression for the energy stored in the capacitor | 1 point |
|-----------------------------------------------------------------------------------|---------|

$$U = \frac{1}{2} CV^2 = \left(\frac{1}{2}\right)(3.23 \times 10^{-6} \text{ F})(4 \text{ V})^2$$

$$U = 2.58 \times 10^{-5} \text{ J}$$

Note: Linear regression gives  $U = 2.56 \times 10^{-5} \text{ J}$

- iii. 2 points



- |                                                                                      |         |
|--------------------------------------------------------------------------------------|---------|
| For a concave down curve starting at the origin and above the original curve         | 1 point |
| For a curve that asymptotes later than the original and at twice the original value. | 1 point |

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**2016 SCORING GUIDELINES**

**Question 3**

**15 points total**

**Distribution  
of points**

(a) 2 points

For correctly using Ohm's law to calculate the current through the resistors

1 point

$$I = \frac{V}{R} = \frac{(20 \text{ V})}{(200 \, \Omega)} = 0.10 \text{ A}$$

Use Ohm's law to calculate the voltage across  $R_2$

$$V = IR = (0.10 \text{ A})(100 \, \Omega)$$

For a correct answer

1 point

$$V = 10 \text{ V}$$

(b) 2 points

For correctly using the equation for the change in current in an inductor

1 point

$$\mathcal{E} = -L \frac{di}{dt}$$

$$\frac{di}{dt} = -\frac{\mathcal{E}}{L}$$

Note: The minus sign may be omitted since the question asks for magnitude

For correctly using the voltage across  $R_2$  to calculate the rate of change of current

1 point

$$\frac{di}{dt} = \frac{(10 \text{ V})}{(3.0 \text{ H})}$$

$$\frac{di}{dt} = 3.3 \text{ A/s}$$

(c)

i. 2 points

For correctly using Ohm's law to calculate the voltage across  $R_2$

1 point

$$V = IR = (0.020 \text{ A})(100 \, \Omega) = 2.0 \text{ V}$$

Use Ohm's law to calculate the current through  $R_3$

$$I = \frac{\mathcal{E} - V_2}{R_1 + R_3} = \frac{(20 \text{ V} - 2.0 \text{ V})}{(50 \, \Omega + 50 \, \Omega)}$$

For a correct answer

1 point

$$I = 0.18 \text{ A}$$

ii. 2 points

For using the equation to calculate the energy stored in the inductor

1 point

$$U = \frac{1}{2} LI^2 = \left(\frac{1}{2}\right)(3.0 \text{ H})(0.18 \text{ A} - 0.020 \text{ A})^2$$

For an answer consistent with part (c)i

1 point

$$U = 0.0384 \text{ J}$$

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**Question 3 (continued)**

**Distribution  
of points**

(d) 2 points

For correctly using Ohm's law to determine the current through the inductor

1 point

$$I_L = \frac{\mathcal{E}}{R_1 + R_3} = \frac{(20 \text{ V})}{(50 \, \Omega + 50 \, \Omega)}$$

For a correct answer

1 point

$$I_L = 0.20 \text{ A}$$

(e) 2 points

For using the loop rule to relate the inductor emf to the voltage drop across  $R_2$

1 point

$$\mathcal{E}_L = V_{R_2}$$

For correctly expressing the loop equation as a differential equation

1 point

$$-L \frac{di}{dt} = iR_2$$

(f) 3 points

For separation of variables in the differential equation above

1 point

$$-L \frac{di}{dt} = iR_2$$

$$\frac{1}{i} di = -\frac{R_2}{L} dt$$

For attempting to integrate with the correct limits or the correct constant of integration

1 point

$$\int_{i_0}^{i(t)} \frac{1}{i} di = \int_{t=0}^{t'=t} -\frac{R_2}{L} dt'$$

$$[\ln(i)]_{i_0}^{i(t)} = \left[ -\frac{R_2 t'}{L} \right]_{t'=0}^{t'=t}$$

$$\ln\left(\frac{i(t)}{i_0}\right) = -\frac{R_2}{L}(t - 0)$$

$$i(t) = i_0 e^{-R_2 t/L} = (0.20 \text{ A}) e^{-(100 \, \Omega)t/(3.0 \text{ H})}$$

For a correct answer

1 point

$$i(t) = (0.20 \text{ A}) e^{-33.3t}$$